StreamingWAM: Streaming World Action Modeling via Chunk-Wise Staircase Flow Matching

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Project page: renshaojie233.github.io/StreamingWAM

Abstract

World Action Models (WAMs), often built upon pretrained video-generation world models, have emerged as a promising framework for general-purpose and open-ended robot control. However, their real-world deployment often suffers from discontinuous and jerky execution. Existing methods typically address this issue by either improving inference efficiency or enhancing action continuity across prediction chunks, without jointly addressing both challenges. To bridge this gap, we propose StreamingWAM, a streaming-diffusion framework for WAM-based robot control. StreamingWAM combines a noise ladder for chunk-wise action prediction, a prefix rollout module for cross-chunk continuity, and dense video-history memory for temporally consistent conditioning. Experiments on LIBERO, LIBERO-Plus, RoboTwin 2.0, and real-world robotic tasks demonstrate that StreamingWAM improves action continuity across static and dynamic settings, achieving competitive performance while reducing steady-state replanning latency to 72.9 ms in simulation and 57.3 ms on the real robot. The method also produces smoother action transitions, while the streaming staircase reduces sensitivity to the execution horizon K.

1 Introduction

Recent WAMs are built upon world models, particularly pretrained video-generation models (Ye et al., 2026b; Li et al., 2026b; Yuan et al., 2026; Kim et al., 2026). Through pretraining objectives based on future-state prediction, these models acquire knowledge of environment dynamics, which is then adapted to robotic domains via fine-tuning for general-purpose and open-ended robot control (Zhang et al., 2026; Wang et al., 2026a). By leveraging this learned dynamics prior, WAMs can exhibit strong generalization when deployed on real-world robots across diverse tasks and environments (Wang et al., 2026b).

Despite their impressive performance, WAMs often exhibit temporally discontinuous and jerky motions during closed-loop execution (MotuBrain Team, 2026; Black et al., 2025). This behavior is closely tied to their non-streaming inference procedure: at each replan, existing WAMs regenerate a complete action chunk from scratch through multi-step denoising and discard the unexecuted suffix of the previous plan. This procedure creates two coupled limitations. First, the combination of a large video-generation backbone, joint video–action processing, and iterative denoising makes WAM inference computationally expensive. Regenerating a complete action chunk at every replan further compounds this cost, resulting in substantial response latency and slower adaptation to new observations (Lu et al., 2026; Ma et al., 2026; Cai et al., 2026; Li et al., 2026a). Second, independently generated chunks may be inconsistent with previously executed actions, causing abrupt transitions at replanning boundaries (Lu et al., 2026; Høeg et al., 2024). Existing methods typically improve either inference efficiency or cross-chunk action continuity, but rarely both within a unified framework.

To bridge this gap, we propose StreamingWAM (Figure 1), which introduces a streaming diffusion paradigm into WAMs. Specifically, the **noise ladder** staggers future action blocks across noise levels in a persistent buffer, enabling low-latency, single-step steady-state replanning. The **prefix rollout module** conditions future-action denoising on a clean prefix of recently executed actions, improving

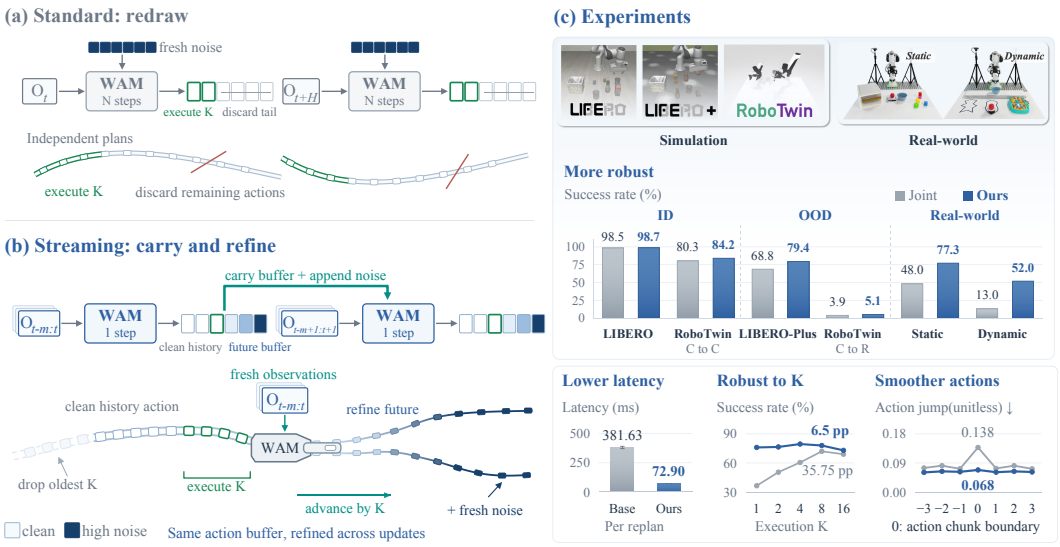


Figure 1: Overview of our streaming world-action model. (a) Standard policies regenerate a complete action chunk at every replan. (b) StreamingWAM rolls the action buffer forward with one denoising step per replan. (c) StreamingWAM performs strongly across simulation and real-world benchmarks.

continuity across action chunks. Finally, **dense video-history memory** provides recent visual context for action prediction, supporting temporally consistent responses to state changes.

We evaluate StreamingWAM on LIBERO, LIBERO-PLUS, RoboTwin 2.0, and real-world manipulation tasks spanning static and dynamic settings, where it achieves competitive success while reducing steady-state replanning latency to 72.9 ms in simulation and 57.3 ms on the real robot. Further analysis shows that the proposed components jointly improve cross-chunk action continuity, reduce sensitivity to the execution horizon K, and support adaptation to changing observations.

Our contributions are summarized as follows:

- We introduce StreamingWAM, which integrates a noise ladder for chunk-wise streaming action prediction, a prefix rollout module for maintaining continuity with previously executed actions, and dense video-history memory for incorporating historical visual context into action prediction. Together, these components improve inference efficiency and execution continuity.
- We evaluate StreamingWAM on LIBERO, LIBERO-PLUS, RoboTwin 2.0, and real-world manipulation tasks spanning both static and dynamic settings. StreamingWAM achieves competitive task success while reducing steady-state replanning latency to 72.9 ms in simulation and 57.3 ms on the real robot. Extensive ablations further validate the contribution of each proposed component.
- Through controlled ablations and static and dynamic evaluations, we analyze the complementary effects of the proposed components. The noise ladder and prefix rollout improve cross-call action continuity, reduce sensitivity to the execution horizon K, and enable low-latency replanning. The clean action prefix and dense video history provide recent action–observation context, supporting temporally consistent execution and adaptation to object motion.

2 Related Work

World action models and vision-language-action models Two dominant paradigms have emerged for generalist robot control. Vision-language-action models (VLAs) such as π_0 (Black et al., 2024) and $\pi_{0.5}$ (Physical Intelligence et al., 2025) build on pretrained vision-language models, combining Internet-scale semantic knowledge with large-scale cross-embodiment robot training. WAMs instead build on pretrained video-generation models and jointly predict future visual states and actions, transferring video-derived dynamics priors to robot control. DreamZero (Ye et al., 2026b) and LingBot-VA (Li et al., 2026b) demonstrated the viability of this paradigm, achieving strong general-

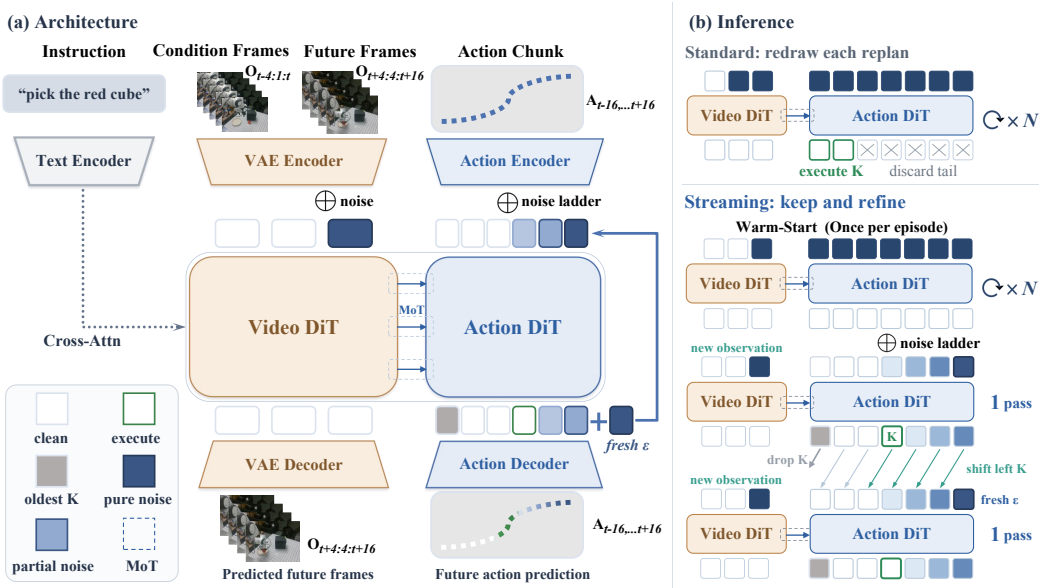


Figure 2: Overview of StreamingWAM. (a) Joint video–action architecture with a clean action prefix and staircase denoising. (b) Standard multi-step inference versus single-step streaming inference.

ization across environments and embodiments. However, large video backbones and iterative joint video–action denoising introduce substantial inference latency, limiting high-frequency closed-loop control.

Efficient world action models Recent work accelerates WAM inference through lightweight action modules, reduced video-side computation, asynchronous video–action denoising, and action-centered architectures (Yuan et al., 2026; Ma et al., 2026; Li et al., 2026a; Guo et al., 2026; Ye et al., 2026a). These methods mainly reduce per-step computation while retaining the standard full-chunk, multi-step replanning loop; in contrast, StreamingWAM carries and refines a persistent action buffer with a single forward pass per replan.

Diffusion policies and streaming diffusion Diffusion Policy generates action chunks through iterative denoising (Chi et al., 2023), whereas Streaming Diffusion Policy (SDP) and Diffusion Forcing use position- or token-dependent noise levels to support rolling and flexible-horizon generation (Høeg et al., 2024; Chen et al., 2024). StreamingWAM adapts this principle to WAMs, preserving the video-conditioned dynamics prior while enabling persistent, single-pass replanning.

3 Method

3.1 Problem formulation

At each planning step t, the robot queries the policy with the current visual observation o_t and task instruction l to obtain an H-step chunk of future actions:

$$a_{t:t+H-1} = \pi(o_t, l). \tag{1}$$

Here, H denotes the future-action prediction horizon. A WAM realizes this policy by jointly denoising the action chunk and a short future video sequence $v_{t:t+T}$, where T is the video prediction horizon. At deployment, an execution horizon $K \leq H$ determines how many predicted actions are applied before replanning: the robot executes $a_{t:t+K-1}$, discards the unused suffix $a_{t+K:t+H-1}$, and replans when the next observation becomes available. For block-aligned streaming deployment, we use execution widths that divide H, ensuring that each roll advances the buffer by an integer number of complete noise-ladder blocks.

Streaming policy formulation. To enable streaming action control, we augment this standard WAM interface with temporal context and a persistent plan buffer. At each replanning time t, the model consumes the m most recent visual observations, the last P actions executed by the robot, and the unfinished plan carried over from the previous call. It returns the next K executable actions together with an updated plan buffer for the subsequent call:

$$(a_{t:t+K-1}, \mathcal{B}_{t+K}) = \pi_\theta \left(\underbrace{o_{t-m+1:t}}_{\text{recent visual observations}}, l, \underbrace{a_{t-P:t-1}}_{\text{executed-action history}}, \underbrace{\mathcal{B}_t}_{\text{unfinished plan}} \right). \tag{2}$$

Here, t indexes environment steps, P is the executed-action history length, and $\mathcal{B}_t$ denotes the persistent plan buffer aligned with the current control step. The robot executes only $a_{t:t+K-1}$. Although the model maintains an H-step action plan internally, action slots beyond the executable prefix remain in $\mathcal{B}_{t+K}$ and are carried forward to the next planning call. These future actions may still be partially denoised and are never issued to the robot until they enter the executable prefix.

Standard WAM inference lacks cross-call plan persistence. At the n-th planning call, a standard WAM initializes a new H-step action chunk from independent noise and runs the complete denoising procedure:

$$a_{1:H}^{(n)} = \mathcal{D}_\theta \left(\epsilon_{1:H}^{(n)}; z_n \right), \qquad \epsilon_{1:H}^{(n)} \sim \mathcal{N}(0, I), \tag{3}$$

where $z_n = z(o_{t_n}, l)$, t_n is the environment time of planning call n, and $\mathcal{D}_\theta$ denotes the complete denoising procedure. Neither the noise initialization nor the intermediate denoising state is passed to call $n+1$. After executing the first K actions, the unexecuted suffix is discarded. Because all action slots follow the same denoising schedule, the model neither reuses the previous plan nor maintains partially refined future actions across calls. The streaming interface in Equation (2) therefore requires a persistent plan buffer and a position-dependent denoising schedule.

3.2 StreamingWAM

Overview We propose StreamingWAM (Figure 2), a streaming diffusion formulation built on a Mixture-of-Transformers (MoT) WAM with dedicated video and action experts. It combines a **prefix rollout module** that conditions prediction on clean executed actions, a **noise ladder** that progressively refines a persistent future-action buffer, and **dense video-history memory** that supplies recent visual context. Together, these components amortize denoising across planning calls, reducing steady-state latency while improving cross-chunk continuity. StreamingWAM requires neither additional trainable parameters nor additional training data.

Prefix rollout module and dense video-history memory At each planning step t, we form an action sequence consisting of a clean prefix of P previously executed actions, followed by H future action slots. The prefix is not a prediction target: during training, it is populated with ground-truth actions preceding the target chunk, whereas at deployment, it contains the actions actually executed by the robot. These prefix actions are provided without noise, allowing the action branch to model the future action sequence as

$$p_\theta(a_{t:t+H-1} \mid z(o_{t-m+1:t}, l),\, a_{t-P:t-1}). \tag{4}$$

Here, $o_{t-m+1:t}$ forms the dense video-history memory, capturing short-term visual dynamics that complement the executed-action prefix to improve the temporal consistency of action generation. As planning advances, newly executed actions are rolled into the conditioning prefix, while the unexecuted portion of $\mathcal{B}_t$ is shifted forward and reused. Thus, overlapping future actions are not regenerated from scratch, breaking the cross-call independence of standard WAM inference described in Section 3.1. Following Black et al. (2025), the action prefix also provides longer-term context than the short visual-history window.

Streaming denoising over the future segment The clean prefix alone does not carry the future plan across calls, since the H future action slots would still be reinitialized from pure noise. We therefore assign them a staircase of noise levels. The future slots are grouped into consecutive blocks of width K, corresponding to the execution horizon, with all positions in each block sharing one noise level and later blocks assigned higher levels. Accordingly, after every K actions are executed, the prefix and the carried plan are rolled forward by exactly one block. Writing $c(j) = \lfloor (j-1)/K \rfloor$ for the block index of the j-th future position and $C = \lceil H/K \rceil$ for the number of blocks, the noise level at that position is

$$\tau_j = \frac{1}{N}\left(\left\lfloor \frac{N\,(c(j)+1)}{C} \right\rfloor - \delta\right), \qquad \delta \sim \mathcal{U}\{0, \ldots, \lfloor N/C \rfloor - 1\}, \tag{5}$$

where N is the number of noise levels. We use $N \geq C$, which makes the offset set nonempty; the implementation additionally clamps discrete noise indices to the valid range for robustness. The random offset δ shifts the whole staircase down, exposing the model to the partially denoised states that occur midway through rolling execution.

Multi-width staircase training. The staircase is entirely determined by the block width K, and we do not hold it fixed. For each training example, we independently sample the execution horizon K uniformly from a finite candidate set $\mathcal{K}$ and construct the corresponding noise profile. We retain the standard shared-timestep objective for a fraction of training steps to preserve the model's multi-step denoising capability, which is required during the warm-up phase at the beginning of each episode, when the initial plan is generated from pure noise.

Training objective Both the video and action branches are trained with weighted flow matching. Given a clean target y, Gaussian noise $\epsilon \sim \mathcal{N}(0, I)$, and a noise level $\tau \in [0, 1]$, we construct the interpolant $y_\tau = (1-\tau)y + \tau\epsilon$ and regress its velocity $\epsilon - y$:

$$\mathcal{L}_{vid} = \mathbb{E}_{y,\epsilon,\tau}\left[w(\tau)\,\|f_\theta(y_\tau, \tau, o, l) - (\epsilon - y)\|_2^2\right], \tag{6}$$

where $w(\tau)$ is inherited from the base model and emphasizes intermediate noise levels. Under ordinary shared-timestep action training, all H future slots use the same scalar τ, so the weight is shared across the suffix:

$$\mathcal{L}_{\text{act}}^{\text{shared}} = \mathbb{E}\left[w(\tau)\frac{1}{H}\sum_{j=1}^{H}\left\|u_j^{\text{shared}} - \left(\epsilon_j - a_j^{\text{fut}}\right)\right\|_2^2\right]. \tag{7}$$

Here, u_j^{shared} is the predicted velocity for future slot j under the shared suffix timestep. Our noise ladder replaces this scalar with a future-slot profile $\boldsymbol{\tau} = (\tau_1, \ldots, \tau_H)$ defined by Equation (5). The action prefix a^{pre} is provided as clean conditioning and excluded from the loss. The full token-wise timestep input is $\bar{\boldsymbol{\tau}} = (\mathbf{0}_P, \boldsymbol{\tau})$. The staircase action objective is

$$\mathcal{L}_{\text{act}} = \mathbb{E}\left[\frac{1}{H}\sum_{j=1}^{H} w(\tau_j)\left\|u_j^{\text{fut}} - \left(\epsilon_j - a_j^{\text{fut}}\right)\right\|_2^2\right], \qquad u^{\text{fut}} = f_\theta\left(a^{\text{pre}}, a_{\boldsymbol{\tau}}^{\text{fut}}, \bar{\boldsymbol{\tau}}, o, l\right). \tag{8}$$

Here, a_j^{fut} and u_j^{fut} denote the j-th future action slot and its predicted velocity, respectively. Since τ_j varies across noise-ladder blocks, $w(\tau_j)$ produces a corresponding block-wise loss profile. Compared with Equation (7), the position-dependent weight therefore moves inside the sum over future slots. The video objective remains unchanged, and the total loss is $\mathcal{L} = \mathcal{L}_{\text{act}} + \lambda\mathcal{L}_{\text{vid}}$, where λ is the relative video-loss weight. We use equal video and action weights, i.e., $\lambda = 1$, in all experiments.

3.3 Inference Process

At deployment, the future plan is maintained in a persistent buffer $\mathcal{B}_t$ of H action slots. At each planning call, a single joint forward pass performs one Euler update toward zero noise. Inference

uses the deterministic zero-offset ladder (i.e., $\delta = 0$ in Equation (5)). For future slot j, we define the pre- and post-update noise levels and their difference as

$$\tau_j^{\text{pre}} = \frac{1}{N}\left\lfloor \frac{N(c(j)+1)}{C} \right\rfloor, \qquad \tau_j^{\text{post}} = \frac{1}{N}\left\lfloor \frac{Nc(j)}{C} \right\rfloor, \qquad \Delta_j = \tau_j^{\text{pre}} - \tau_j^{\text{post}}. \tag{9}$$

For our evaluated settings, C divides N, so $\Delta_j = 1/C$ for every future slot. Writing $\boldsymbol{\tau}^{\text{pre}} = (\tau_1^{\text{pre}}, \ldots, \tau_H^{\text{pre}})$ and $\boldsymbol{\Delta} = (\Delta_1, \ldots, \Delta_H)$, the buffer is updated and rolled as

$$\begin{aligned} \widetilde{\mathcal{B}}_t = \mathcal{B}_t - \boldsymbol{\Delta} \odot f_\theta^{\text{act}}(\mathcal{B}_t, \boldsymbol{\tau}^{\text{pre}}, o_{t-m+1:t}, l, a_{t-P:t-1}), \qquad a_{t:t+K-1} = \widetilde{\mathcal{B}}_{t;1:K}, \\ \mathcal{B}_{t+K} = \text{Concat}\left(\widetilde{\mathcal{B}}_{t;K+1:H}, \epsilon^{\text{new}}\right). \end{aligned} \tag{10}$$

Here, $\widetilde{\mathcal{B}}_t$ is the buffer after the Euler update, and $\epsilon^{\text{new}} \sim \mathcal{N}(0, I)$ is a fresh block of K noisy actions. The leading block reaches zero noise and is executed, while the remaining $H - K$ actions are retained and shifted forward. The executed block is added to the clean action prefix for the next planning call but is not stored in $\mathcal{B}_{t+K}$.

Warm-up initialization. At the beginning of each episode, we perform a one-time warm-up to construct the initial persistent buffer $\mathcal{B}_0$. Specifically, we first generate a clean action chunk using the standard multi-step flow-matching sampler and then re-noise its blocks according to the staircase, following the denoise initialization strategy of Høeg et al. (2024). Since no executed-action history is available at this point, the action prefix a^{pre} is initialized with zeros. After this warm-up, each planning call follows the prefix-rollout procedure and requires only a single joint forward pass.

4 Experiments

4.1 Setup

Benchmarks and baselines. We conduct experiments on LIBERO (Liu et al., 2023) and LIBERO-PLUS (Fei et al., 2025). LIBERO comprises four 10-task suites (OBJECT, SPATIAL, LIBERO-10, and GOAL), while LIBERO-PLUS extends these tasks to 10,030 perturbed evaluation instances, enabling systematic evaluation under distribution shifts. We further evaluate on ROBOTWIN 2.0 (Chen et al., 2025), a dual-arm manipulation benchmark with clean and domain-randomized settings, to assess whether the findings transfer across benchmarks and embodiments.

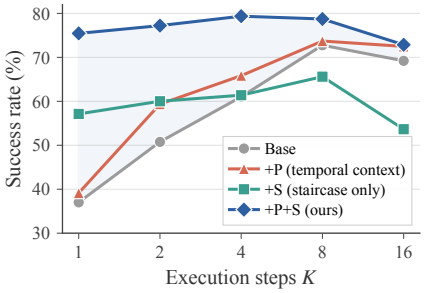


Figure 3: Execution-horizon sweep on the full LIBERO-PLUS evaluation.

Finally, we conduct experiments on both static and dynamic real-world manipulation tasks using a Franka Research 3 arm equipped with a Robotiq 2F-85 gripper, two external Intel RealSense D435i cameras, and a wrist-mounted ZED Mini camera (Section 4.3). We compare StreamingWAM against 13 competitive baselines spanning both VLA policies and WAMs, grouped according to whether they use embodied data during pretraining. Detailed baseline descriptions are provided in Section A.4.

Metrics. For most tasks, we report success rate (%); for Whack-a-Mole, we report valid activations of the currently lit button within a fixed time budget. We distinguish the once-per-episode time to first action (TTFA), which includes buffer initialization, from the steady-state replanning latency used during normal closed-loop control (Lu et al., 2026). Continuity metrics include per-step position change, joint displacement from the initial posture, and a discrete acceleration proxy.

Implementation details Following Fast-WAM (Yuan et al., 2026), we build our WAM on Wan2.2-TI2V-5B. The action-expert backbone is initialized by linearly interpolating the pretrained video-DiT

weights to its dimensions and applying α-scaling to preserve their magnitude, while the action-specific encoder and output head are randomly initialized. At inference, $\mathcal{B}_0$ is initialized using a 10-step flow-matching sampler and re-noised according to the staircase. Each subsequent planning call uses a single Euler update. Detailed training and inference configurations, including the number of optimization and denoising steps, action and execution horizons, are provided in Section A. For controlled training-budget comparisons, Fast-WAM-Joint and StreamingWAM start from the same benchmark-specific Joint checkpoint and receive the same number of additional optimisation steps under the Joint and streaming objectives, respectively. The main-table LIBERO-PLUS baseline instead follows the official FastWAM-Joint evaluation configuration, as stated in the table caption.

Table 1: Success rates (%) on LIBERO and LIBERO-PLUS. The final column reports success pooled over all LIBERO-PLUS evaluation instances. FastWAM-Joint follows its official evaluation configuration; detailed inference settings are provided in Table 6. EP: embodied pre-training. **Bold**: best per column; underline: second best.

			Perturbation type							
Method	EP	LIBERO	Cam.	Robot	Lang.	Light	Bkgnd.	Noise	Layout	Overall
With embodied pre-training										
π0.5	✓	96.9	**75.4**	77.5	85.6	96.9	94.6	89.7	**85.7**	**85.7**
GE-ACT	✓	96.5	60.7	77.0	77.4	95.8	86.0	**90.9**	80.2	80.3
VLA-JEPA	✓	97.2	63.3	67.1	85.4	95.6	93.6	66.3	85.1	79.5
X-VLA	✓	98.1	23.4	**89.7**	75.7	88.2	**96.0**	62.7	71.8	71.4
OpenVLA-OFT	✓	97.1	56.4	31.9	79.5	88.7	93.3	75.8	74.2	69.6
π0-Fast	✓	85.5	65.1	21.6	61.0	73.2	73.2	74.4	68.8	61.6
π0	✓	94.2	13.8	6.0	58.8	85.0	81.4	79.0	68.9	53.6
UniVLA	✓	95.2	1.8	46.2	69.6	69.0	81.0	21.2	31.9	42.9
Without embodied pre-training										
Faster-WAM	–	98.0	67.9	49.0	92.1	94.3	57.0	82.3	82.7	75.0
4D-WAM	–	98.4	45.2	64.3	90.6	94.3	57.7	69.1	79.2	71.0
Fast-WAM-Joint	–	98.5	42.5	60.8	**92.9**	91.8	55.4	63.3	78.4	68.8
WorldVLA	–	79.1	0.1	27.9	41.6	43.7	17.1	10.9	38.0	25.0
Fast-WAM	–	97.6	16.4	44.5	68.9	78.2	53.7	37.7	60.7	51.5
Ours	–	**98.7**	61.0	79.0	92.0	**97.8**	64.7	78.6	83.6	79.4

4.2 Main results

Table 1 shows that StreamingWAM achieves the strongest overall performance among WAMs without embodied-data pretraining, reaching 98.7% on LIBERO and 79.4% on LIBERO-PLUS. The gain is most pronounced under distribution shift, where it outperforms the closest architectural baseline, Fast-WAM-Joint, by 10.6 percentage points on LIBERO-PLUS, particularly under camera-viewpoint and robot-initial-state perturbations. This comparison uses each method's reported deployment configuration (Table 6) and is a system-level comparison rather than a matched-K ablation.

For RoboTwin 2.0 (Table 2), task-specific training uses the same 2,500 clean demonstrations. StreamingWAM reaches 84.2% clean and 5.1% randomized success, compared with 80.3% and 3.9% for Fast-WAM-Joint. The gap to embodied-pretrained $\pi_{0.5}$ highlights the value of cross-embodiment data for out-of-distribution generalization.

Table 2: RoboTwin 2.0 success rate (%). EP: embodied pre-training; bold/underline: best/second best.

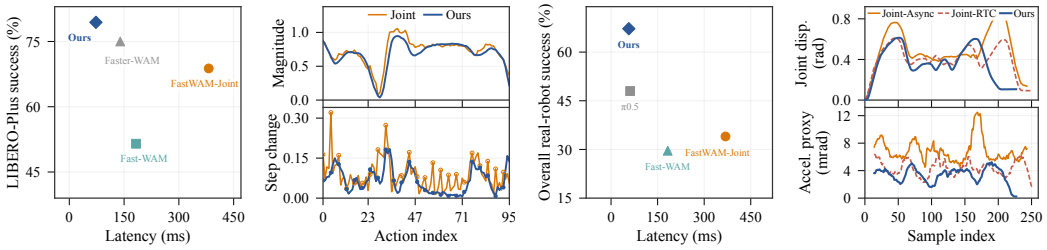
Method	EP	Clean	Random	Avg.
With embodied pre-training				
Spatial Forcing	✓	77.2	9.5	43.4
π0.5	✓	70.7	**46.0**	**58.4**
X-WAM	✓	70.0	25.8	47.9
X-VLA	✓	68.0	20.9	44.5
EventVLA	✓	65.6	15.7	40.7
Xiaomi-Robotics-0	✓	62.9	18.2	40.6
G0	✓	62.7	9.1	35.9
Abot-M0	✓	57.4	22.9	40.2
Without embodied pre-training				
Fast-WAM-Joint	–	80.3	3.9	42.1
Fast-WAM	–	77.8	1.9	39.9
AHA-WAM	–	64.3	3.2	33.8
Ours	–	**84.2**	5.1	44.7

Inference cost We report episode-initial TTFA separately from steady-state replanning latency. StreamingWAM takes 72.9 ms in simulation and 57.3 ms on the real robot during normal steady-state control (Figure 4), corresponding to 5.2× and 6.4× speedups over the standard 10-step

Fast-WAM-Joint deployment. These system-level comparisons include different horizons and evaluation counts. One real-robot update is 57.3 ms versus 46.8 ms for a timing-only one-step Joint call; the speedup comes from amortizing denoising so each steady replan uses one model evaluation instead of ten. The one-time episode-initial warm start is reported separately in Table 7.

Action continuity StreamingWAM produces smoother actions in both simulation and real-world execution. On LIBERO-PLUS, the boundary-to-interior jump ratio drops from 1.840 to 1.105 and global translation-command jerk decreases by 59.0% relative to Fast-WAM-Joint (Figure 4b and Table 8).

Real-robot trajectories show the same trend, with smoother joint displacement and a lower discrete-acceleration proxy (Figure 4(d)).



(a) Simulation efficiency. (b) Simulation continuity. (c) Real-robot efficiency. (d) Real-robot continuity.

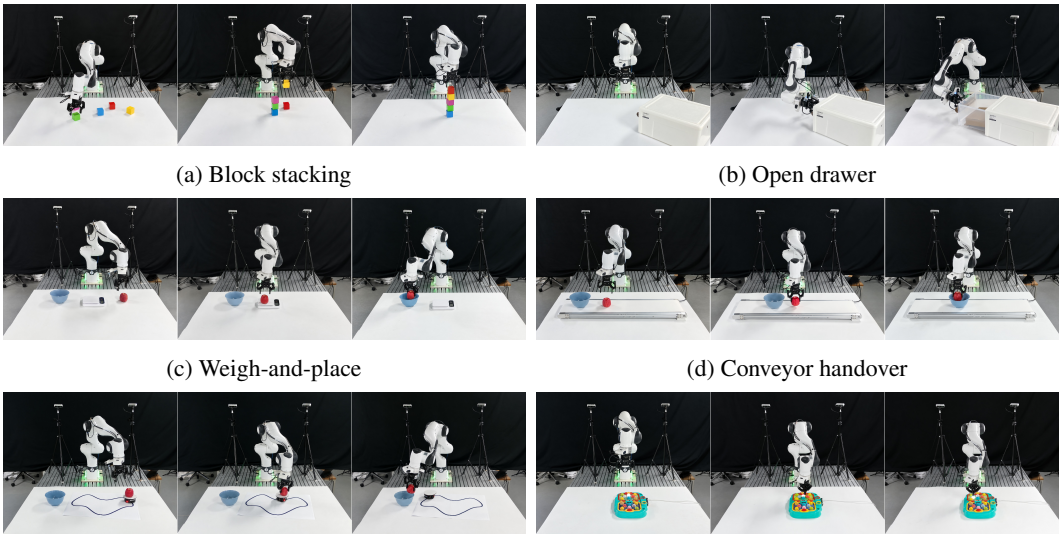
Figure 4: Inference efficiency and action continuity in simulation and on the real robot. Panel (c) is a system-level summary: for Fast-WAM and FastWAM-Joint, the latency coordinate is the standard 10-step latency, while overall success uses the task-specific real-robot deployment in which dynamic tasks use one denoising step.

4.3 Real robot experiments

Setup We deploy on a Franka Research 3 arm with a Robotiq 2F-85 gripper, two external Intel RealSense D435i cameras, and a wrist-mounted ZED Mini following the DROID (Khazatsky et al., 2024) configuration. The control loop runs at 15 Hz on an RTX 5090 workstation. Demonstrations are collected via GELLO joint-space teleoperation; details are provided in Section E. On the deployment GPU, steady-state per-replan latency is 57.3 ms for our model (1 denoise step) and 61.5 ms for $\pi_{0.5}$ (10 steps), both within the 66.7 ms control budget. Fast-WAM and FastWAM-Joint require 182.6 ms and 368.6 ms, respectively, with their standard 10-step inference, missing multiple deadlines per replan (Table 7). We therefore use 10 steps on static tasks and one step on dynamic tasks for Fast-WAM and FastWAM-Joint; preliminary 10-step dynamic deployments yielded near-zero success because the policy could not react within the task time scale. The six evaluation tasks are illustrated in Figure 5, with detailed descriptions provided in Section E.1.

Results Figure 6 shows that StreamingWAM performs best overall. It achieves the highest aggregate success across the five manipulation tasks (168/250, 67.2%, Wilson 95% CI [61.2, 72.7], versus 120/250, 48.0% [41.9, 54.2], for the strongest baseline), leads both dynamic tasks (32/50 on conveyor handover and 20/50 on car pursuit, compared with baseline maxima of 15/50 and 8/50), and records the most whack-a-mole presses (424 vs. 408).

Analysis Two complementary temporal effects may explain the task-level gains. First, the largest margins occur on the dynamic tasks. A single frame locates an object but generally leaves its velocity ambiguous, whereas consecutive real observations reveal frame-to-frame motion. This temporal evidence may allow the WAM's future-state predictor to anticipate object motion before action selection. At the same time, single-step streaming replanning lowers the steady-state latency, allowing those predictions to be refreshed at the task's reaction time scale. The dynamic gains therefore plausibly reflect the joint effect of motion evidence from real observation history, the WAM's predictive prior, and low-latency replanning, rather than any one factor alone. In stage-matched conveyor trials,



(a) Block stacking (b) Open drawer

(c) Weigh-and-place (d) Conveyor handover

(e) Car pursuit (f) Whack-a-mole

Figure 5: Real-robot tasks. (a)–(c) Static; (d)–(f) dynamic. Each panel shows one successful rollout.

FastWAM-Joint largely freezes the apple, whereas StreamingWAM predicts displacement along the conveyor direction. The corresponding qualitative future-video prediction sequence and displacement measurements are provided in the appendix (Figure 9). Because the examples come from independent trials and the models differ in more than their visual context, they provide qualitative mechanism evidence rather than a controlled history-only ablation. Second, temporal context may also help track progress in multi-stage tasks. On weigh-and-place, StreamingWAM succeeds in 42/50 trials versus 25/50 for the strongest baseline. Recent observations together with the executed-action history provide short-horizon context about whether the weighing stage has been completed and whether the apple should next be transferred to the bowl. This interpretation is consistent with the task-level gain, although it remains to be isolated by a task-specific context ablation.

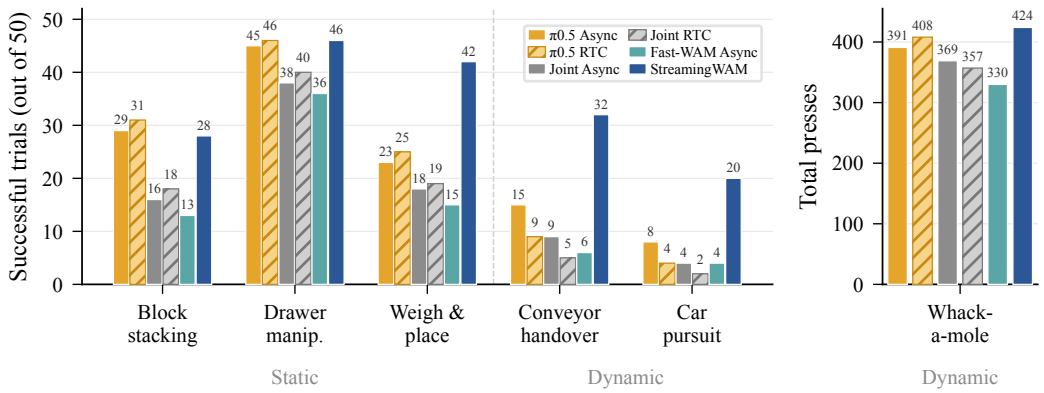


Figure 6: Real-robot results. Each of the five manipulation tasks (left) reports the number of successful trials out of 50; whack-a-mole (right) reports total valid lit-button activations across 50×30 s sessions.

4.4 Ablation studies and analysis

In this section, we ablate the key components of StreamingWAM and conduct further analyses of its behavior, yielding the following findings:

(1) The staircase flattens the performance curve across execution horizons, and the combined temporal-context package raises its overall level. Figure 3 and Table 3 compares the clean

Table 3: Ablation studies on the full 10,030-episode LIBERO-PLUS evaluation (success rate, %). The top block compares the temporal-context package (P) and streaming staircase (S); the bottom block varies one design choice from the full model. Spread is max−min across the reported K values.

Variant	K=1	2	4	8	16	Spread
Base (Fast-WAM-Joint)	37.01	50.77	61.03	72.76	69.21	35.75
+ Temporal context (P)	39.14	59.36	65.83	73.71	72.49	34.57
+ Staircase (S)	57.13	60.01	61.39	65.60	53.65	11.95
+ P + S (full model)	**75.45**	**77.22**	**79.36**	**78.73**	**72.86**	**6.50**
w/ single-width K=4	71.52	74.08	75.31	73.74	70.39	4.92
w/ sparse history	56.35	67.80	69.86	68.45	60.43	13.51
w/ uniform loss weight	75.52	78.37	78.11	77.50	71.27	7.10
w/o warm start	70.30	72.86	74.21	75.23	72.56	4.93

action prefix bundled with dense visual history (P) and the streaming staircase (S) across execution horizons. The staircase reduces Spread from 35.75 pp for Base and 34.57 pp for +P to 11.95 pp for +S; combining P and S further reduces it to 6.50 pp while achieving the highest success at every evaluated K. Thus S provides K-robustness, while the bundled temporal context raises performance; finding (3) separately tests history density.

(2) Multi-width staircase training improves performance across execution horizons. As shown in Table 3, replacing the full training set $\mathcal{K}$ with a fixed staircase width of $K = 4$ preserves low variation across execution horizons but reduces success at every evaluated value of K, including its training width (75.31% versus 79.36% at $K = 4$). The fixed-width model peaks at $K = 4$, as expected, but generalizes less effectively to other execution horizons. These results support the design motivation in Section 3.2: sampling multiple widths exposes the model to more diverse staircase geometries and improves performance across the entire K-sweep.

(3) Dense visual history improves success rates and reduces variation across execution horizons. With P and S held fixed, replacing dense visual history (stride 1) with sparse history (stride 4) shifts the entire K-sweep downward and nearly doubles the Spread (Table 3). The dense frames supply recent visual context that overlaps with the clean-prefix window, giving the model a more accurate estimate of the current scene state. The benefit is most pronounced at small K, where frequent replanning places greater importance on recent observations.

(4) Auxiliary training choices mainly affect accuracy rather than horizon robustness. Uniform loss weighting stays within 1.59 pp of the full model across all K and changes Spread only from 6.50 to 7.10 pp, indicating that position-dependent weighting is a modest refinement. On the full evaluation, removing the warm start reduces the $K = 4$ result by 5.15 pp (74.21% versus 79.36%). This shows that initialization mainly raises the initial buffer's accuracy; the staircase remains the mechanism tested for robustness across execution horizons.

(5) The streaming advantage vanishes when the execution width spans the full prediction horizon. The training noise profiles sample $\mathcal{K}_{\text{train}} = \{1, 2, 4, 8\}$, all block-aligned divisors of $H = 16$, while $K = 16$ is held out from training. At $K = H = 16$, the horizon becomes one block and loses the inter-block staggering required for streaming. The full model then drops by 5.87 pp from $K = 8$ to 16 (78.73% to 72.86%), while its margin over the temporal-context-only (+P) variant shrinks from 13.53 pp at $K = 4$ to 0.37 pp at $K = 16$. Thus, this boundary stress test shows that the streaming benefit vanishes when the execution width spans the prediction horizon.

5 Limitations

Our study has two main limitations. First, our experiments use a single base-model family, Fast-WAM. Although the streaming formulation does not require a specific WAM backbone, its effectiveness

on other backbones remains to be validated. Second, our model does not use large-scale embodied pre-training. On RoboTwin 2.0's randomized-initialization split, embodied-pretrained policies such as $\pi_{0.5}$ (Physical Intelligence et al., 2025) retain substantially higher success under scene variation. This gap may partly reflect the broader data coverage and stronger generalization provided by cross-embodiment pre-training. Combining such pre-training with streaming denoising is an important direction for future work.

6 Conclusion

We presented StreamingWAM, a streaming-diffusion framework that amortizes action denoising across planning calls through a persistent rolling buffer. Its streaming staircase progressively refines future actions, while the executed-action prefix and dense visual history maintain temporal context across replans. Across LIBERO-Plus, RoboTwin 2.0, and real-robot tasks, StreamingWAM achieves competitive task success, reduces steady-state replanning latency to 72.9 ms in simulation and 57.3 ms on the real robot, improves cross-chunk action continuity, and compresses execution-horizon sensitivity from 35.75 to 6.50 percentage points. Its largest real-robot gains appear on moving-target and multi-stage tasks, consistent with the value of temporal context and responsive replanning. Together, these results support streaming action generation as a practical approach to more responsive and continuous WAM control.

Code and data availability

Code, model checkpoints, the real-world dataset, and supplementary videos are available from the project page.

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A Reproducibility details

A.1 Model configuration

We use the same dual-expert Mixture-of-Transformers backbone in all three domains. The Video and Action DiTs contain 30 aligned blocks. In every block, the two experts form their own queries, keys, and values, perform joint mixed self-attention, and then return to expert-specific cross-attention and feed-forward paths. The Action DiT residual width is 1,024, whereas its 24 heads use an internal QKV width of $24 \times 128 = 3{,}072$; these are different quantities.

Table 4: Core model configuration by benchmark. Video/Action values are shown as V/A.

Benchmark	Layers	Hidden dim.	FFN dim.	Heads × dim.	Act./prop.	Cameras	Resolution ($H \times W$)
LIBERO / LIBERO-PLUS	30/30	3,072/1,024	14,336/4,096	24×128	7/8	2, horiz.	224×448
RoboTwin 2.0	30/30	3,072/1,024	14,336/4,096	24×128	14/14	3, top + 2 bottom	384×320
Real robot	30/30	3,072/1,024	14,336/4,096	24×128	8/8	3, vertical	384×224

A.2 Training configuration

For each direct comparison, the baseline and StreamingWAM share a Joint initialisation and receive the same number of additional optimisation steps. The baseline retains ordinary shared-timestep Joint training, whereas StreamingWAM uses the streaming objective. The resulting schedules are listed in Table 5.

For LIBERO and LIBERO-PLUS, the second-stage branches use AdamW with learning rate 10^{-4}, betas $(0.9, 0.95)$, $\epsilon = 10^{-8}$, weight decay 10^{-2}, gradient clipping at 1.0, bf16 precision, and a 60,000-step cosine schedule without warm-up. We report the checkpoint at step 40,000, selected for evaluation. For StreamingWAM, 80% of optimization minibatches use the staircase objective with $\mathcal{K}_{\text{train}} = \{1, 2, 4, 8\}$, 32 noise levels, and a random staircase offset; the remaining 20% use ordinary shared-timestep flow matching. The video and action objectives have equal weight, the clean 16-step action prefix is conditioning only, and the future horizon is $H = 16$. We additionally evaluate the held-out, block-aligned width $K = 16$.

Table 5: Training schedule by benchmark.

Benchmark	Method	Shared Joint stage	Second stage	Total steps	Global batch
LIBERO / LIBERO-PLUS	Fast-WAM-Joint	34,720	+40,000 Joint	74,720	80→64
	StreamingWAM	34,720	+40,000 streaming	74,720	80→64
RoboTwin 2.0	Fast-WAM-Joint	135,776	+135,776 Joint	271,552	64→64
	StreamingWAM	135,776	+135,776 streaming	271,552	64→64
Real robot	Fast-WAM-Joint	20,000	+30,000 Joint	50,000	128→128
	StreamingWAM	20,000	+30,000 streaming	50,000	128→128

The FastWAM-Joint rows use ordinary shared-timestep flow matching, whereas the StreamingWAM rows use the 80/20 mixture above. All other methods retain the training values reported in their original papers—including their reported optimization budget, batch size, and learning-rate settings—rather than being silently retrained with our recipe. In particular, Fast-WAM follows Yuan et al. (2026), Faster-WAM follows Ma et al. (2026), and the VLA baselines follow their cited original implementations. We do not infer an unreported hyperparameter.

A.3 Evaluation and inference configuration

LIBERO-Plus results use the full 10,030-episode protocol (OBJECT 2,518 / SPATIAL 2,402 / LIBERO-10 2,519 / GOAL 2,591). Success is determined by the environment's built-in criterion with no tolerance modification. RoboTwin 2.0 results use all 50 tasks with seed 0 under both clean and randomised initialisations, following the official leaderboard protocol. Real-robot demonstrations are collected with a GELLO joint-level teleoperation device at 15 Hz. The training set contains 590

Table 6: Inference hyperparameters used in our evaluations. H is the predicted future-action horizon (the 16 clean prefix actions of StreamingWAM are excluded), and K is the number of actions executed per replan. $10 \rightarrow 1$ denotes a one-time 10-step warm start followed by one rolling SDP update per replan. For real-robot non-streaming rows, $10/1$ means static/dynamic tasks.

	LIBERO / LIBERO-Plus			RoboTwin 2.0			Real robot		
Method	Steps	H	K	Steps	H	K	Steps	H	K
$\pi_{0.5}$ Async	10	10	5	10	50	50	10	16	4
$\pi_{0.5}$ RTC	10	10	5	10	50	50	10	16	4
Fast-WAM	10	32	10	10	32	24	$10/1$	32	4
Faster-WAM	10	32	4	10	32	4	10	32	4
Fast-WAM-Joint Async	10	32	10	10	32	24	$10/1$	32	4
Fast-WAM-Joint RTC	10	32	4	10	32	4	10	32	4
StreamingWAM	$10 \rightarrow 1$	16	4	$10 \rightarrow 1$	16	8	$10 \rightarrow 1$	16	4

trajectories across seven collection tasks: block stacking, open drawer, close drawer, weigh-and-place, conveyor handover, car pursuit, and whack-a-mole. Open and close drawer are collected separately but belong to the same manipulation family. Evaluation reports the open-drawer variant and omits the easier close-drawer variant, yielding the six evaluation tasks described below. The number of trajectories varies by task, with some tasks containing fewer than 100 demonstrations. Evaluation uses 50 trials per manipulation task and 50×30 s sessions for whack-a-mole. Within $\pi_{0.5}$ and FastWAM-Joint, Async and RTC variants share a checkpoint and differ only in execution strategy.

Introduced by Lu et al. (2026), TTFA measures the wall-clock time from an episode's initial observation to its first executable action at the controller. For StreamingWAM, this once-per-episode quantity includes the warm start that initializes the persistent action buffer. Normal closed-loop control does not repeat that cost, so our primary latency metric is steady-state replanning time; the initialization latency is reported separately. All measurements use the same RTX 5090 GPU with CUDA-synchronous timing.

During simulation evaluation, StreamingWAM uses synchronous control: the environment waits for each policy call before executing the returned K-action block. Real-robot deployment instead uses asynchronous control. For $K = 4$, the next inference is launched only after the first three actions of the current block have finished, so inference overlaps with execution of the fourth action.

The $\pi_{0.5}$ simulation entries use the public implementation defaults: LIBERO predicts 10 actions and executes five per query, while RoboTwin predicts and executes 50; both use 10 flow-matching integration steps. Async and RTC therefore share the same simulation defaults and differ only in the real-robot execution stack.

The RoboTwin Joint setting executes $K = 24$ actions, whereas StreamingWAM uses $K = 8$; the latter value is an execution setting and must not be confused with the training set $\mathcal{K}_{\text{train}}$. The real-robot FastWAM-Joint RTC row is the configuration reported in Figure 6 and Table 12. The FastWAM-Joint LIBERO and LIBERO-PLUS result in Table 1 follows its official configuration with $K = 10$. StreamingWAM's steady-state one is a rolling update of a persistent buffer, not one-step denoising of an independently redrawn Joint chunk.

A.4 Baselines

The LIBERO comparison includes 13 baselines spanning both VLA policies and WAMs, grouped according to whether they use embodied data during pretraining. Eight methods use embodied-data pretraining: π0.5 (Physical Intelligence et al., 2025), GE-ACT (Liao et al., 2025), VLA-JEPA (Sun et al., 2026), X-VLA (Zheng et al., 2025), OpenVLA-OFT (Kim et al., 2025), π0-Fast (Pertsch et al., 2025), π0 (Black et al., 2024), and UniVLA (Bu et al., 2025). The remaining five do not: Faster-WAM (Ma et al., 2026), 4D-WAM (Yang et al., 2026), Fast-WAM-Joint, WorldVLA (Cen et al., 2025), Fast-WAM (Yuan et al., 2026). The RoboTwin comparison additionally includes AHA-WAM (Cai et al., 2026). Among these, Fast-WAM-Joint, the jointly trained variant of Fast-WAM, serves as our closest direct baseline because it shares the same model architecture with StreamingWAM while omitting the three modules introduced in Section 3.2.

B Extended simulation results

B.1 Inference latency

Table 7 lists the steady-state per-replan latency for all open-source WAM variants measured on the same RTX 5090 GPU with CUDA synchronous timing. Baseline architectures are reproduced from the published code with random weights (network structure determines latency, not learned weights); our model uses the exact checkpoint reported in the paper.

Table 7: Inference latency comparison. All measurements on the same RTX 5090 with CUDA synchronous timing. n: number of timed forward passes.

Method	H	Steps	n	Latency (ms)	LIBERO-PLUS (%)
Fast-WAM-Joint	32	10	30	381.6 ± 6.9	68.8
Fast-WAM	32	10	30	182.6 ± 3.2	51.5
Faster-WAM	32	10	30	139.4 ± 1.7	75.0
Ours (steady)	16	1	300	**72.9** ± 0.3	**79.4**
Ours (initial warm start)	16	10	30	406.2 ± 0.8	—
Real-robot deployment (same RTX 5090)					
Fast-WAM-Joint	32	10	30	368.6 ± 2.4	—
Fast-WAM-Joint (timing only)	32	1	30	46.8 ± 0.3	—
Fast-WAM	32	10	30	182.6 ± 3.2	—
$\pi_{0.5}$	16	10	30	61.5 ± 1.2	—
Ours (steady)	16	1	300	**57.3** ± 0.6	—
Ours (initial warm start)	16	10	30	436.8 ± 1.6	—

The 1-step FastWAM-Joint timing row uses the same checkpoint as the 10-step row. One Streaming-WAM forward is not cheaper than one Joint forward (57.3 versus 46.8 ms). Its steady-state advantage comes from the persistent buffer, which reduces each replan from ten model evaluations to one. Because the reported deployment configurations also differ in H, these speedups are system-level rather than a controlled single-variable comparison. In the real-robot task results, Fast-WAM and FastWAM-Joint use 10 steps on static tasks and one step in the dynamic-task main setting, as summarized in Table 6. The one-step dynamic setting is necessary because preliminary 10-step deployments yield near-zero success at the required reaction time scale.

B.2 Action continuity

Table 8 reports the action-continuity metrics underlying Figure 4b. All three models are evaluated with 1-step inference and $K{=}4$ on a 500-episode balanced subset of LIBERO-PLUS. Translation and rotation jumps are the ℓ_2 norm of the end-effector command change at replanning boundaries; the boundary-to-interior ratio measures the ratio of the jump at step 0 to the average jump at non-boundary steps; global translation jerk is the mean ℓ_2 norm of the finite- differenced acceleration of end-effector translation commands.

Table 8: Action-continuity metrics on LIBERO-PLUS (500-episode subset, 1-step inference, $K{=}4$). $\downarrow$: lower is better.

Method	SR (%)	Trans. jump $\downarrow$	Rot. jump $\downarrow$	Bnd./int. ratio $\downarrow$	Jerk $\downarrow$
Fast-WAM	60.2	0.1350	0.02903	1.831	5.775
Fast-WAM-Joint	68.8	0.1379	0.02681	1.840	5.710
Ours	**79.0**	**0.0684**	**0.00879**	**1.105**	**2.341**
Ours vs. Fast-WAM-Joint		-50.4%	-67.2%	—	-59.0%

B.3 Per-suite breakdown on LIBERO-PLUS

Table 9 breaks down the success rate by suite and perturbation type for Fast-WAM-Joint (Joint) and our model. Our model improves on every suite, with the largest gains on OBJECT (+16.4 pp)

and LIBERO-10 (+13.8 pp). Across perturbation types, the gap is widest under camera-viewpoint (+18.5 pp pooled) and robot-initial-state (+18.2 pp) perturbations, where Fast-WAM-Joint is most vulnerable.

Table 9: Per-suite per-perturbation success rate (%) on LIBERO-PLUS (10,030 episodes). Ours uses $K=4$; Joint denotes the official-configuration Fast-WAM-Joint baseline from Table 1. BG: background textures; Robot: robot initial states; Cam: camera viewpoints; Lang: language instructions; Noise: sensor noise; Obj: objects layout; Light: light conditions.

Suite		BG	Robot	Cam	Lang	Noise	Obj	Light	All
OBJECT	Joint	64.1	57.8	49.2	97.7	68.0	77.4	97.3	72.2
	Ours	83.9	82.2	83.6	97.7	92.2	83.9	98.0	**88.6**
SPATIAL	Joint	74.0	66.3	55.6	93.3	72.1	90.4	99.0	78.5
	Ours	73.6	83.7	64.4	97.4	87.5	96.4	97.9	**86.1**
LIBERO-10	Joint	32.2	72.3	30.8	96.6	64.1	87.8	84.7	66.3
	Ours	53.6	88.3	54.7	100.0	76.6	92.9	98.5	**80.1**
GOAL	Joint	54.4	48.2	35.8	84.9	48.8	61.6	85.3	59.0
	Ours	50.9	62.8	42.6	74.4	57.5	64.9	96.8	**63.4**
Pooled	Joint	55.4	60.8	42.5	92.9	63.3	78.4	91.8	68.8
	Ours	64.7	79.0	61.0	92.0	78.6	83.6	97.8	**79.4**

C Per-task results on RoboTwin 2.0

Table 10 reports the per-task success rates of StreamingWAM under the clean and randomized object-pose settings. Averaging the 50 tasks gives 84.2% and 5.1%, respectively, matching the aggregate results in Table 2.

Table 10: Per-task success rate (%) of StreamingWAM on all 50 RoboTwin 2.0 tasks. The table is arranged in two blocks for compactness.

Task	Clean	Randomized	Task	Clean	Randomized
adjust_bottle	100.0	2.0	place_can_basket	60.0	6.0
beat_block_hammer	93.0	1.0	place_cans_plasticbox	96.0	6.0
blocks_ranking_rgb	99.0	2.0	place_container_plate	99.0	4.0
blocks_ranking_size	45.0	0.0	place_dual_shoes	86.0	2.0
click_alarmclock	100.0	40.0	place_empty_cup	91.0	5.0
click_bell	100.0	17.0	place_fan	80.0	0.0
dump_bin_bigbin	93.0	1.0	place_mouse_pad	54.0	0.0
grab_roller	98.0	5.0	place_object_basket	87.0	3.0
handover_block	98.0	0.0	place_object_scale	81.0	0.0
handover_mic	99.0	3.0	place_object_stand	94.0	7.0
hanging_mug	18.0	0.0	place_phone_stand	83.0	0.0
lift_pot	94.0	4.0	place_shoe	94.0	5.0
move_can_pot	85.0	0.0	press_stapler	84.0	3.0
move_pillbottle_pad	95.0	1.0	put_bottles_dustbin	91.0	0.0
move_playingcard_away	96.0	0.0	put_object_cabinet	51.0	1.0
move_stapler_pad	42.0	0.0	rotate_qrcode	86.0	0.0
open_laptop	85.0	0.0	scan_object	87.0	4.0
open_microwave	64.0	5.0	shake_bottle	100.0	28.0
pick_diverse_bottles	75.0	9.0	shake_bottle_horizontally	100.0	22.0
pick_dual_bottles	85.0	11.0	stack_blocks_three	95.0	0.0
place_a2b_left	95.0	3.0	stack_blocks_two	99.0	12.0
place_a2b_right	90.0	1.0	stack_bowls_three	83.0	9.0
place_bread_basket	89.0	13.0	stack_bowls_two	90.0	4.0
place_bread_skillet	90.0	7.0	stamp_seal	65.0	0.0
place_burger_fries	92.0	8.0	turn_switch	54.0	1.0
Mean	**84.2**	**5.1**			

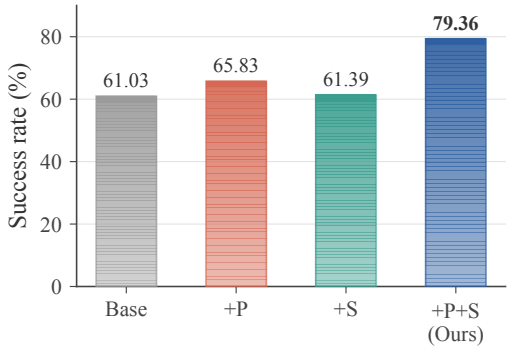


Figure 7: Component contributions on LIBERO-PLUS at $K = 4$. The clean action prefix with dense visual history (P) and the streaming staircase (S) have complementary effects and combine to achieve the strongest result.

D Additional ablation results

D.1 Component contributions at $K = 4$

D.2 Warm-start ablation

Section 4.4 reports the aggregate effect of warm-start length. Table 11 gives the per-setting breakdown on the full 10,030-episode evaluation: removing the warm start drops success by 5.2 pp, while a single step already recovers the gap.

Table 11: Warm-start ablation on LIBERO-PLUS at K=4.

Warm-start steps	Successes	Total	Success (%)
0	7,443	10,030	74.2
1	7,934	10,030	79.1
4	7,919	10,030	79.0
8	7,927	10,030	79.0
10	7,960	10,030	79.4

E Real-robot experiment details

E.1 Task descriptions

We evaluate six manipulation tasks, grouped by whether the target object moves during execution. *Static tasks*: **block stacking**—stack five layers of blocks; **open drawer**—grasp the handle and pull the drawer open; and **weigh-and-place**—place an apple on a kitchen scale, read the weight, then transfer it to a bowl. *Dynamic tasks*: **conveyor handover**—pick a moving apple off a conveyor belt and place it into a bowl; **car pursuit**—intercept an apple riding a curved-path toy car and place it into a bowl; and **whack-a-mole**—press the randomly lit button on a response-time toy. The dynamic tasks require the policy to continuously adapt its actions to moving objects and therefore directly evaluate reactive control in time-varying environments.

E.2 Observation preprocessing

Figure 8 shows a representative model input frame from the real robot. Each of the three camera views (two exterior RealSense D435i and one wrist-mounted ZED Mini) is resized to 128×224 and vertically stacked into a 384×224 tensor (all dimensions are $H\times W$). In simulation, LIBERO provides two views horizontally concatenated into a 224×448 tensor.

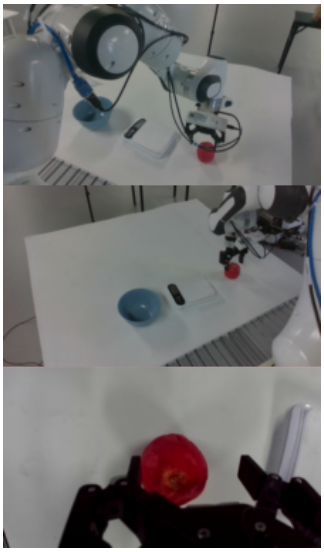


Figure 8: Real-robot model input: three camera views (exterior 1, exterior 2, wrist) vertically concatenated into a 384×224 frame ($H \times W$).

E.3 Action continuity on the real robot

Figure 4(d) compares aligned real-robot trajectory segments from FastWAM-Joint under asynchronous and RTC execution with StreamingWAM. The upper trace reports joint displacement from the initial posture, while the lower trace reports the supplied discrete acceleration proxy. StreamingWAM remains lower than both Joint baselines for most samples; we do not interpret the proxy as physical acceleration or jerk.

E.4 Qualitative moving-target anticipation

To further examine why the largest real-robot gains occur on dynamic tasks, Figure 9 visualizes future video predictions before contact in stage-matched conveyor trials. FastWAM-Joint largely freezes the apple, whereas StreamingWAM predicts displacement along the conveyor direction. The examples come from independent trials and the two models differ in more than their visual context, so this evidence supports an anticipation interpretation but does not isolate history as the sole cause.

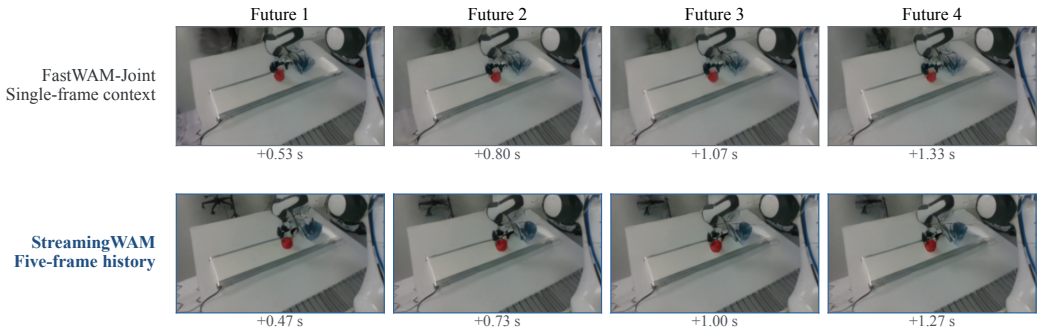


Figure 9: Qualitative moving-target prediction before contact. Four future frames are replayed offline from recorded conveyor-task inputs and action chunks. FastWAM-Joint uses single-frame context; StreamingWAM uses five-frame dense history. In external camera 3, the predicted/recorded apple displacement is 1.38/6.93 pixels for FastWAM-Joint and 9.66/5.93 pixels for StreamingWAM; the cosine between predicted and recorded displacement directions is 0.497 and 0.995, respectively. The examples are stage-matched independent trials, so they support a qualitative anticipation interpretation but do not isolate history as the sole cause.

E.5 Per-task results

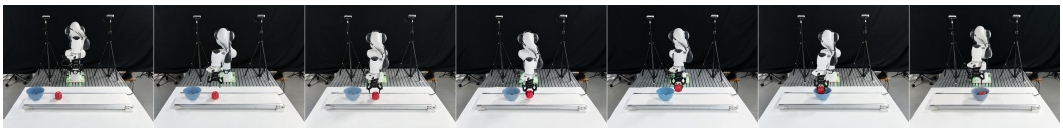
Table 12 reports per-task results for all six real-robot tasks. The five manipulation tasks are evaluated over 50 trials each; whack-a-mole reports valid activations of the currently lit button across 50×30 s sessions. An incorrect or repeated press is a failure; reaching the device's failure limit triggers a reset that consumes part of the session.

Table 12: Real-robot per-task results. Number of successful trials for the five manipulation tasks (50 trials each); valid lit-button activations for whack-a-mole (50×30 s sessions).

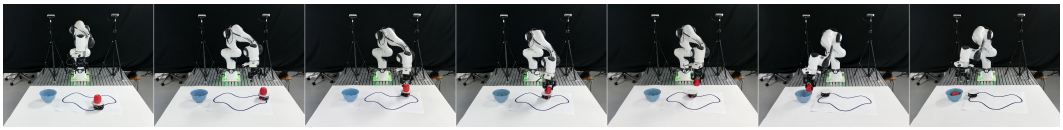
	$\pi_{0.5}$		FastWAM-Joint		Fast-WAM	**StreamingWAM**
Task	Async	RTC	Async	RTC	Async	**Async**
Static tasks						
Block stacking	29	**31**	16	18	13	28
Open drawer	45	**46**	38	40	36	**46**
Weigh-and-place	23	25	18	19	15	**42**
Dynamic tasks						
Conveyor handover	15	9	9	5	6	**32**
Car pursuit	8	4	4	2	4	**20**
Five-task total	120	115	85	84	74	**168**
Whack-a-mole (valid activations)	391	408	369	357	330	**424**

E.6 Task execution keyframes

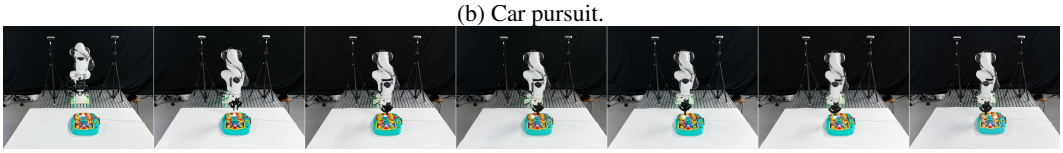
Figures 10a to 10f show successful rollouts of all six real-robot tasks under streaming single-step replanning.



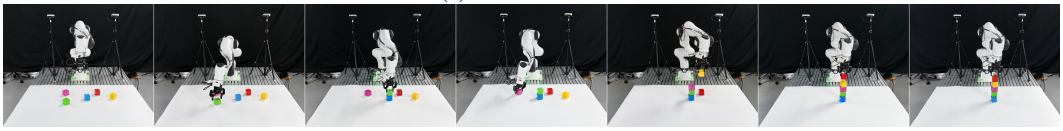
(a) Conveyor-belt handover.



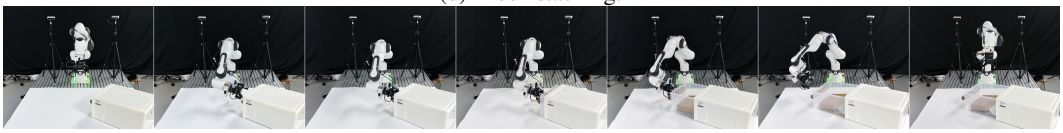
(b) Car pursuit.



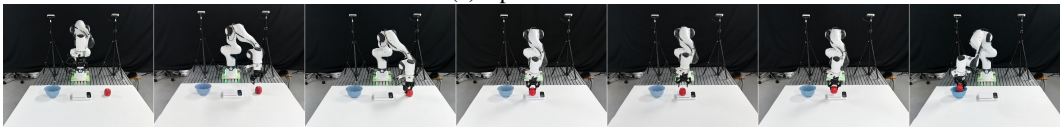
(c) Whack-a-mole.



(d) Block stacking.



(e) Open drawer.



(f) Weigh-and-place.

Figure 10: Real-robot task execution under streaming single-step replanning. Each row shows seven keyframes from one successful episode.